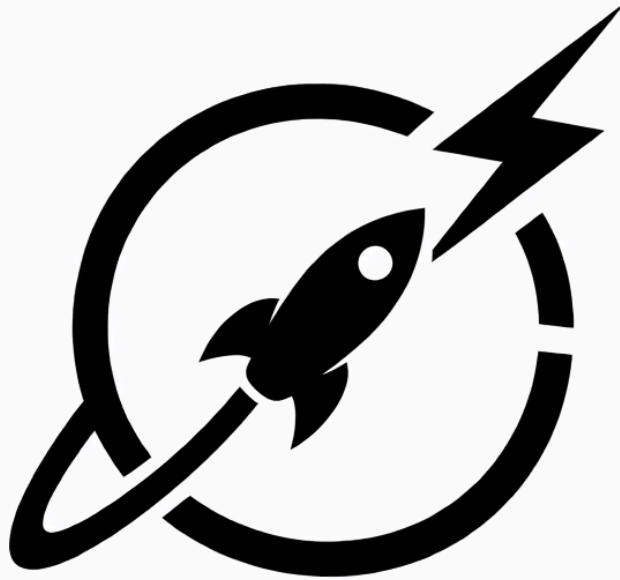


Manufacturing and Testing Review

Electric Dysfunction

Arcs Team 1



**ELECTRIC
DYSFUNCTION**

Contents

1 - Team Structure.....	3
2 - Timeline.....	3
2.1 - Structures.....	3
2.2 - Avionics.....	4
3 - Budget.....	4
4 - Updated Open rocket Simulations.....	5
4.1 - Motor change:.....	5
4.2 - Velocity off rod:.....	5
4.3 - Stability.....	5
4.4 - Ground hit velocity.....	6
4.5 - Apogee.....	6
5 - Manufacture.....	6
5.1 - Fin assembly.....	7
5.2 - Body tube.....	7
5.3 - Nose cone.....	8
5.4 - Bulk heads.....	8
5.5 - Custom PCB Manufacture.....	9
5.6 - Avionics Bay.....	11
5.7 - Decoration.....	12
6 - Structural Testing.....	13
6.1 - Nose cone drop testing.....	13
6.2 - Compression tests.....	13
6.3 - Sag test.....	13
6.4 - Fin test.....	14
7 - Functional testing.....	14
7.1 - Parachute, shock cord and eye bolts.....	14
7.2 - Separation.....	15
7.3 - Avionics Testing.....	15
7.3.1 Tests to be Performed:.....	16
8 - Remaining project risks.....	17
8.1- parachute not deploying.....	17
8.2 - Eye bolts.....	17
8.3 - Recovery.....	17
8.4 - Total Power Loss.....	18
9 - Next steps.....	18
9.1 - Lessons learned from completed testing.....	18
9.2 - Future Testing plans.....	18
9.3 - Future Manufacturing.....	19
10 - References.....	20

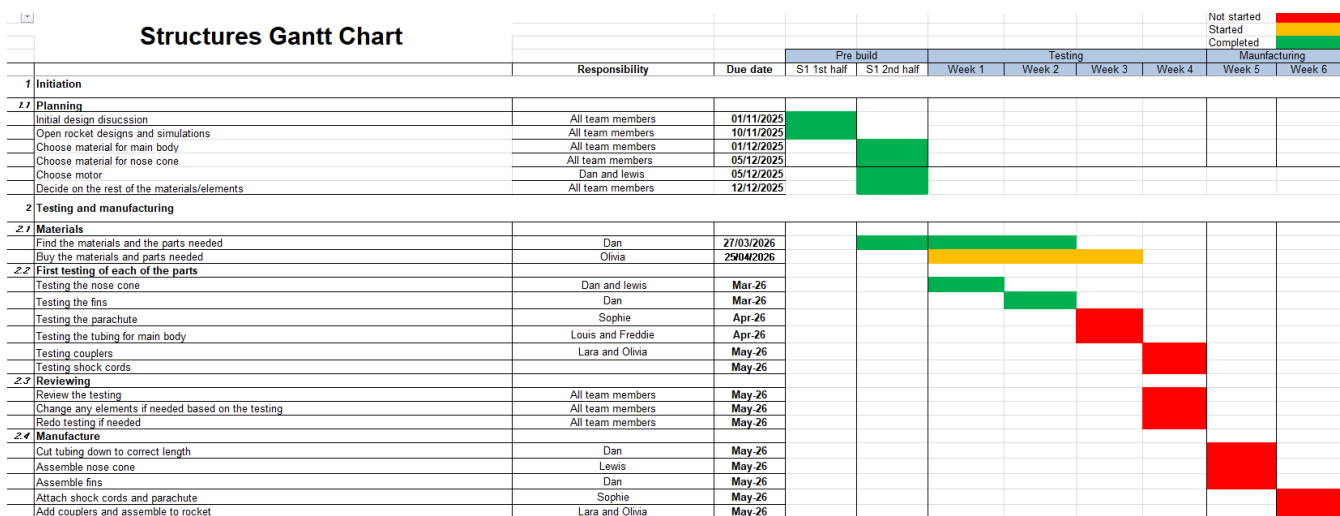
1 - Team Structure

Post the CDR, the structure of the team has stayed the same, however as manufacturing and testing has begun there are new allocated assignments for each member of the team. Each teammate has been assigned a specific task for either testing/manufacturing as listed below.

Name	Role	Lead	Assignment
Sam Orlin	Avionics	Avionics lead	Payload Design and Manufacture
Finn Woolley	Avionics		Data Analysis & Nose Cone R&D
Alistair Wye	Avionics		Data Manipulation & Filtering
Charlie Sims	Avionics		Payload Bay Design
Olivia Gransden	Structures	Team lead	Organisation/testing
Lara Burke	Structures	Vice team lead	Organisation/testing
Louis Goodwin	Structures	Structures lead	Strength testing
Lewis Cairns	Structures		Nose cone design/testing
Sophie Shandley	Structures		Testing/manufacture
Dan Hubbard	Structures	Vice structures lead	Fin R&D/testing and OpenRocket designs
Robert Seddon	Structures		Testing/manufacture
Freddie Kendall	Structures		Strength testing
Tyler Cain	Structures		Testing/manufacture
James Barnet	Structures		Nose cone testing

2 - Timeline

2.1 - Structures



There were many changes to the rocket design once the team was provided with feedback from the CDR, resulting in a small setback with the testing and manufacturing stages, causing the timeline to shift most of the testing to April rather than March as previously intended. The focus now is making sure each element has been tested and changed, if needed, ready for the final assembly.

2.2 - Avionics

Phase	TASK TITLE	TASK LEAD	START DATE	DUE DATE	DURATION (DAYS)	PCT OF TASK COMPLETE	WEEK 1			WEEK 2			WEEK 3			WEEK 4			WEEK 5			WEEK 6			WEEK 7			WEEK 8			WEEK 9		
							M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F		
1	Procurement and Initial Code																																
1.1	All Components Procured	H&P	08/12/25	10/12/25	2	100																											
1.1.1	Core Sensor System Functional	H&P, S	10/12/25	10/12/25	2	100																											
1.2	High-Rate Logging Achieved	S	11/12/25	12/12/25	2	100																											
1.3	Data Pipeline Operational	S, A&V	12/12/25	26/12/25	7	100																											
2	Power, Analysis & Design																																
2.05	All Components Procured	H&P	9/2/26	11/2/26	2	100																											
2.1	Voltage Monitoring Circuit	H&P	11/2/26	13/2/26	2	50																											
2.2	Critical Power Supply Tested	H&P	11/2/26	13/2/26	2	100																											
2.4	Full Analysis Script Complete	A&V, S	9/2/26	23/2/26	14	50																											
2.6	Flight Ready Schematic Finalized	H&P, M&D	16/2/26	18/2/26	2	100																											
	PCB acquired	H&P	15/3/26	3/4/26	19	100																											
3	Manufacturing and Integration																																
3.1	Avionics Board Functional	All Team	30/03/26	02/04/26	2	100																											
3.2	Avionics Bay Complete	M&D	30/03/26	17/04/2007	18	50																											
3.2.1	Full Mission Simulation	All Team	15/04/26	17/04/26	15	15																											
3.3	GPS System Complete	M&D	15/04/26	17/04/26	0	0																											
3.2.2	Launch Readiness	All Team	TBC	TBC	10	10																											

With the core structural components manufactured and the Avionics PCB successfully verified, the focus shifts toward final system integration and software refinement. The upcoming phase will be broken down into the following key objectives:

- **Hardware Integration:** Finalising the integration of the PETG avionics sled into the main airframe, as well as securely mounting and wiring the u-blox M10 GPS telemetry system.
- **Power Monitoring:** Establishing a voltage divider circuit. This will step down the 8.4V main flight battery to a safe logic level, allowing the microcontroller to monitor battery health and log discharge curves during flight.
- **Software & Data Pipeline:** Transitioning into the software testing stage. This phase will focus heavily on implementing sensor filtering algorithms to smooth raw barometric and IMU data for accurate apogee detection and optimising the I2C/SPI data pipeline to ensure high-frequency logging to the MicroSD card.

3 - Budget

	Description	Team	Quantity	Unit cost	Shipping	Total cost	link	status
6	MicroSD Module	Avionics	1	3.4	3.9	7.3	https://thepihut.com/products/adafruit-microsd-module	Received
7	Microcontroller	Avionics	1	19.8	0	19.8	https://thepihut.com/products/arduino-pro-mini	Received
8	Barometer	Avionics	1	10.6	0	10.6	https://thepihut.com/products/adafruit-bme280	Received
9	MicroSD Card	Avionics	1	9	0	9	https://thepihut.com/products/sandisk-ultra	Received
10	IMU	Avionics	1	12.5	0	12.5	https://thepihut.com/products/adafruit-mpu6050	Received
11	LCD	Avionics	1	8.7	0	8.7	https://thepihut.com/products/wavesha	Received
12	Misc Avionics	Avionics	1	12.97	0	12.97	Screenshot 2026-03-10 022748.jpg	Received
13	M3 Threaded Rod	Avionics	1	2.31	0	2.31	https://www.aliexpress.com/item/1005691881201.html	Received
14	Battery	Avionics	1	7.4	2.45	9.85	https://www.hobbyrc.co.uk/qnb-930m	Ordered
15	ABS piping	Struct...	2	14.17		28.34	https://www.plasticpipeshop.co.uk/1-11	Received
16	Nose cone	Struct...	1	0	0	0		Not ordered
17	parachute	Struct...	1	22.13	4.17	26.3	https://modelrockets.co.uk/scratch-build-parachute	Ordered
18	shock cord - inextensible	Struct...	1	2	5.24	7.24	https://modelrockets.co.uk/ukroc-26-c	Ordered
19	shock cord - extendable	Struct...	1	4.3		4.3	https://modelrockets.co.uk/scratch-build-shock-cord	Ordered
20	eyebolts	Struct...	1	4.29		4.29	https://www.amazon.co.uk/PATIKIL-M	Not ordered
21	spray paint	Misc	1	9.99		9.99	https://www.amazon.co.uk/Fluorescen	Not ordered
22	rail guides	Struct...	1	8.1		8.1	https://modelrockets.co.uk/scratch-build-rail-guides	Not ordered
23	epoxy glue	Struct...	1	20.99		20.99	https://modelrockets.co.uk/ukroc-26-c	Received
24	epoxy resin	Struct...	1			0		Not ordered
25	motor retainers	Struct...	1	3.4		3.4	https://modelrockets.co.uk/scratch-build-motor-retainers	Not ordered
26	coupler	Struct...	1	4.79		4.79	https://modelrockets.co.uk/scratch-build-coupler	Not ordered
27	motor sleeve	Struct...	1	2.12			https://modelrockets.co.uk/scratch-build-motor-sleeve	Not ordered
28								
29								

The spreadsheet above displays the current costing sheet for all components necessary at this point in the competition. The approximate total cost is currently £206, slightly exceeding the reimbursement of £200 from ARCS, however members of the team have contributed to make up for any current and potential future additional expenses that may arise during manufacturing and testing. Some aspects of the manufacturing do not have any cost due to the use of university facilities such as the iForge, where student credit and free facilities/materials will be used where possible. Now that research and costing for parts has been finalised, the components that are linked in the costing sheet are in the process of being ordered and tested by the team.

4 - Updated Open rocket Simulations

As before, one goal for the team was to reach a maximum apogee of 610m while attempting to keep the flight of the rocket as stable as possible as well as safe to fly; the rocket must also hit the ground at a speed which means it will be recoverable. The simulation took place with a wind speed of 10 knots to give a realistic understanding of the rocket's performance.

Wa...	Name	Configuration	Velocity off rod	Apogee	Velocity at ...	Optimum delay	Max. velocity	Max. accelera...	Time to apogee	Flight time	Ground hit vel...
✓	Simulation 13	[G78G4-10]	14 m/s	630 m	10.6 m/s	9.43 s	125 m/s	103 m/s²	11.2 s	115 s	8.07 m/s

Figure 1 - simulation data from open rocket

4.1 - Motor change:

Due to an error on the open rocket which added an additional 2 second delay to the motor, the G87-7 was incorrectly chosen during the design process. This mistake was discovered in the CDR causing a new motor to be required. The G78-7 was swapped for a G78-10 due to its closeness to the optimum delay of 9.43s. The dimensions of each motor are the same causing no changes to the fin assembly to be made.

4.2 - Velocity off rod:

The design has a velocity off rod of 14m/s, slightly slower than previously, but still good enough to ensure the rocket is aerodynamically stable during early parts of the flight including take-off. The fins will allow the rocket to correct itself here.

4.3 - Stability

This design is suitable as it has a stability of 2.16 calibers, meaning the center of pressure is just over two body diameters behind the center of gravity, providing a strong but not excessive restoring force for stable flight. If stability was too much lower than this, then the rocket wouldn't have a smooth take-off as well as ascent. Especially if there is wind on the day, this would mean the rocket is prone to being blown off course. If the rocket was too stable, it would make it prone to weathercocking so stable, steady flight would once again not be achieved.

4.4 - Ground hit velocity

The estimated ground hit velocity is now slightly higher at 8.07m/s. After testing was carried out on the nose cone, it was established that the nose cone would survive an impact with grass at this velocity and so no further modifications to help this are needed.

Once the descent rate of the rocket is measured with the parachute, it will be compared to the simulated descent rate on OpenRocket. A higher descent rate would increase the likelihood of breaking on landing, while a much lower descent rate would reduce landing accuracy. The rate should ideally fall around 7m/s as it provides a balance between safe touchdown and minimal drift distance.

4.5 - Apogee

An initial assumption was that OpenRocket overestimated the overall apogee by approximately 10%. This led to the initial design having a higher apogee of 674m to allow for this. However it is now led to believe OpenRocket is more accurate so there was too much overcompensation, meaning there were changes in the design to allow the new apogee of 635m to be reached. This will mean the rocket will reach the desired height of 610m, whilst still compensating for any differences.

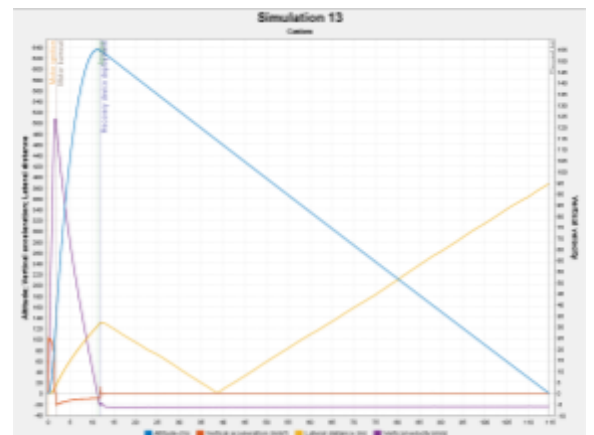


Figure 2 - simulation data from open rocket

The updated model also reaches an estimated lateral distance of approximately 390m, which is better than previous designs as a result of improved stability on the rocket. This will help when trying to recover the rocket.

5 - Manufacture

Manufacture started shortly after the CDR was submitted. The main focus was to manufacture the fin assembly as that is the most complicated and intricate structural part of the rocket. In addition, a large focus was put into the avionics bay manufacture so its connection to the rest of the rocket can be tested.

During manufacture several issues were found with the design causing new designs to be made which differ from the CDR. These include the nose cone, fin assembly and bulkheads. All changes will be explained and justified in their corresponding sections.



Figure 3: rocket assembly CAD

5.1 - Fin assembly

Post CDR, the fin assembly was laser cut and glued. The first iteration used 3mm plywood for both the centering rings and fins.



Figure 5 - Fin assembly V2 drying after being glued

Through inspection, it was decided that the centering rings needed to be thicker so they would be more suitable to withstand the forces of the rocket motor. On the second iteration the centering rings were made of 6mm plywood and the fins 3mm plywood (**see figure 4**). Fin assembly V1 was fit tested with the G78 motor, during this it was discovered that a motor sleeve was necessary to ensure a good fit with the motor. This was then incorporated into the CAD design (**see Figure 6**).

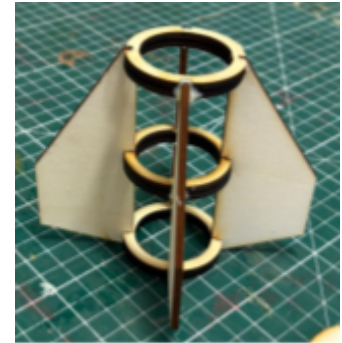


Figure 4 - Fin Assembly V1



Figure 6: Exploded view of the CAD model

In addition to this it was noted that the fin size was quite small in proportion to the length of the rocket. This led to the design of the fin assembly V2 (**see figure 5**). Design V2 has the fins moved up the body of the rocket. Moving the fins up the body reduces the effect on the rocket stability requiring them to become larger to have the same effect. This makes the V2 fins much larger than the original design (approximately double the area). Note that in **figure 5** the assembly is positioned upside down. During the assembly of V2 it was found to be very difficult to align all the fins to be at 90° from each other. Due to

this a Jig was designed to keep the fins straight during gluing. This will be used during the assembly of fins V3, and will be 3D printed out of PLA to ensure it has the correct dimensions.

The fin assembly will be attached to the tubing via four 195mm long slits cut into each side of the tubing. The current plan is to use a coping saw to do this - however alternate methods such as using a dremel or a bandsaw are being considered for their ability to cut straighter lines with more ease. The glued fin assembly can then slide into the tube and be attached using epoxy glue. To assure the cuts are in the correct location a Jig has been designed to assist in the cutting - this will be 3D printed out of PLA to minimise human error and ensure the slits are in the correct places.



Figure 7 - fin assembly testing rig

5.2 - Body tube

Two 1m long ABS tubes with inner diameter 42.7mm and wall thickness 2.8mm were purchased for manufacture and testing. One tube was cut into 2 sections of 730mm and 230mm via a tenon saw. The 730mm length will be used for strength testing (**see section 6.2**). The 230mm length was used to manufacture a mock fin assembly for strength testing

(see section 6.5). Once the strength testing has taken place the 730mm length will be cut into 400mm and 330mm lengths to imitate the dimensions of the rocket. These sections will be fitted with a coupler and be used for the sag test (see section 6.4). The second tube has been left aside to be used for the manufacture of the rocket's components in case any of the tests damage the body tubing.

5.3 - Nose cone

Nose cone V1 was 3D printed in the iForge using PLA. At first the nose cone was printed as one continuous piece to improve its overall strength. However, due to its length it just exceeded the limit for what could be printed in the iForge. This meant it had to be redesigned yet again and printed in 2 sections which was only just allowed to be printed, being close to the maximum time allowed. These 2 halves were then attached together via epoxy glue and left to set. The nose cone was then sanded and used for testing (see section 6.1) to remove any excess epoxy and get a better look and representation of the real thing. It was also done to practice the assembly process.

Due to the new fin V2 design, the nose cone was shrunk to reduce its mass. This created Nose cone V2 (see figure 8). Which has reduced the amount of print that will be used and also the length of the nose cone to decrease the weight.

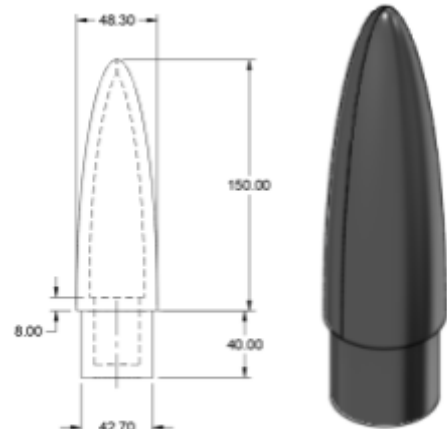


Figure 8: nose cone design 2

The method to attach the nose cone has been changed. Instead of using epoxy glue to hold the nose cone in place, stainless steel pan-head screws will be used. This is to allow the nosecone to be removed so that the avionics bay can be accessed. However, if screws are used the extraction of the avionics bay becomes much simpler and any complications surrounding getting it out are gone. Different screws than what would be used for the bulkheads would be used as this is a plastic compared to wood.

5.4 - Bulk heads

The bulkhead design has also been altered post CDR. After further research into how the rear assembly of a model rocket is designed [1], an additional bulkhead was added behind the original 2 to provide a buffer against the motor itself. Furthermore, the rear 2 bulkheads were redesigned to allow airflow from the charge to pass through and deploy the recovery system effectively. Gaps were placed in

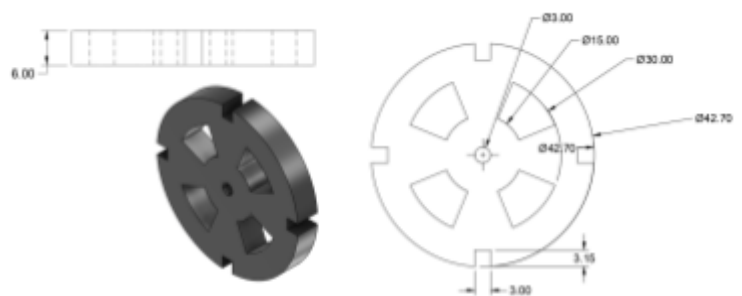


Figure 9 - bulkhead design and dimensions

the design to allow air to pass through the bulkhead after being redirected from their original path - this also meant that the rear bulkheads acted as baffles, protecting the shock cord and parachute from the harmful ejection gases released from the charge.

Another change from the CDR was the decision to use 6mm Plywood rather than 3mm - with this increase in thickness the strength of the bulkhead would be more likely to withstand the hot gas ejection of the motor as well as the forces from the recovery system deployment. Additionally, the increased thickness provides a greater surface area for gluing the bulkheads onto the body tube, minimising the risk of a rapid unplanned disassembly upon parachute deployment.

5.5 - Custom PCB Manufacture

The flight computer is a custom-routed PCB developed over many months of an iterative design process using KiCad. Since the CDR, the PCB design has changed significantly. The previous design did not consider the height added due to the breakout boards and power connectors. Due to the small inner diameter of the rocket, the design was required to be much thinner and more compact. The pull up resistors and decoupling capacitors were removed once it was discovered that the breakout boards already included them internally. Tracing was improved, including parallel connections between sensors, and all components were moved closer together. A custom I2C connector (JST-PH 4 pin) was added to connect a screen for the apogee display. The buzzer system was also removed due to concerns about ineffectiveness and legality raised in the CDR feedback, and replaced with a GPS module and ELRS receiver to transmit GPS coordinates to an ELRS ground station. The battery capacity was increased from 800mah to 1500mah due to concerns about high current draw.

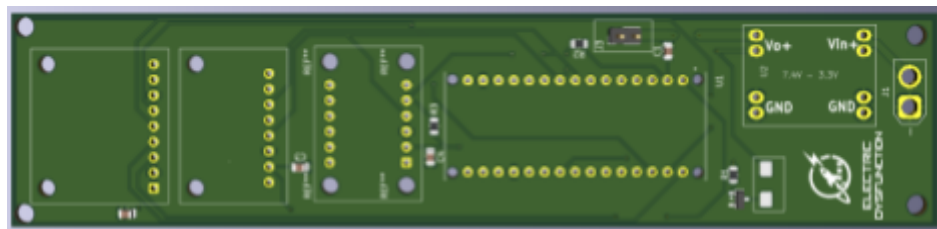


Figure 10 - PCB pre CDR

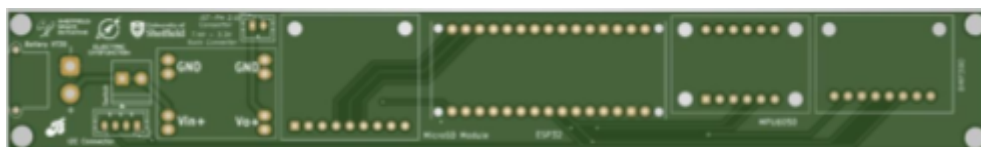


Figure 11 - PCB post CDR

Upon receiving the fabricated boards from the manufacturer, a rigorous unpowered visual and multimeter inspection was conducted. Isolation between ground and live was confirmed for both the 8.4V and 3.3V lines. During this phase, a mistake was identified. The screw

terminals for the external power switch were bridged due to a trace routing error in the KiCad file. This meant that the switch was always closed, regardless of the screw terminal state.

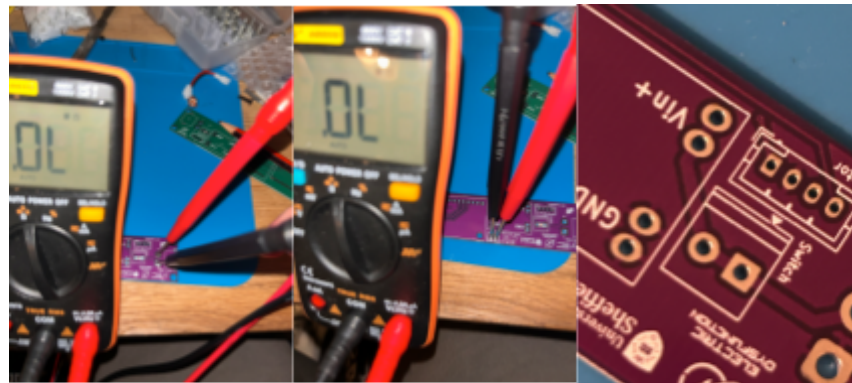


Figure 12 - Testing on Voltage lines

Instead of waiting weeks for a manufacturing replacement, the board was manually reworked using a precision blade to carefully remove the copper trace bridging the pads. Subsequent continuity tests confirmed the bridge was successfully cleared and the circuit was open as intended.

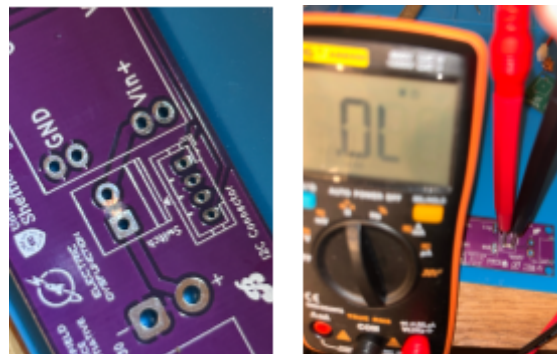


Figure 13 - testing bridge

Following the rework and initial continuity tests, all through hole components were soldered onto the board by hand using lead free solder and high quality flux before beginning the Avionics Testing ([see 7.3](#)). Rubber dampers are added onto the sensors and microSD module to prevent vibrations causing noise or affecting the data logging. M2 x 8mm screws held in place with acrylic hex nuts mount the modules to the board.



Figure 14 - fully assembled PCB

5.6 - Avionics Bay

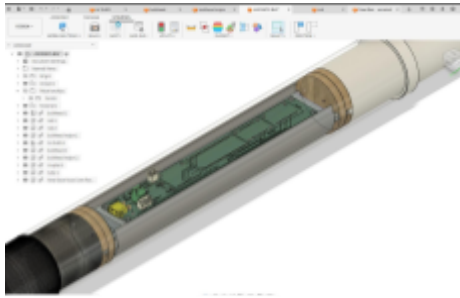


Figure 15 - Avionics bay

The avionics suite is housed within a 3D-printed sled, designed for accessibility and structural rigidity. The assembly is centred around two stainless steel threaded M3 rods that provide the primary load path for the internal components.

The threaded rods are secured using Nyloc nuts and external tooth lock washers. This combination was selected specifically to resist the high frequency vibrations experienced during the motor burn and the mechanical shock of parachute deployment, ensuring

the stack does not loosen in flight.

The internal assembly is suspended between two bulkheads each held in place via external stainless steel pan-head screws. To access the avionics bay, the nose cone is first removed by unscrewing the side entry screws holding it in place. Next the avionics bulkheads are unscrewed, allowing the entire AV bay to slide out.

Version 1: Initial Proof of Concept

The initial sled design served as a proof of concept to verify the board dimensions with the airframe and to visualise the space available. However, testing revealed that the clearance holes for the structural threaded rods were too small, preventing the M3 threaded rods from sliding along the sled. Furthermore, the holes ran too close to the edge of the bay, causing a failed print. This design did not include a dedicated, secure mounting location for the flight battery as the battery dimensions were not known at the time.

Version 2: Refined Tolerances, Heat-Set Inserts and Battery Accommodations

V2 significantly improved upon the mechanical stability of the sled. The holes for the threaded rods were adjusted, using a 3.7mm diameter hole instead of the previous 3mm. This, and changing the position of these holes so that they weren't so close to the edge, eliminated the printing failures of V1. Securing the PCB directly into the plastic print risked thread stripping in the high vibration environments, so a soldering iron was used to embed brass, heat-set inserts into the sled allowing the PCB to be screwed in and out without degrading the plastic. A dedicated battery cavity was also drafted into this model which needs major improvements due to space issues and cable management ineffectiveness.



Figure 16 - V2 sled designs

Version 3:

Version 3 provided improvements on V2, particularly improving on the battery cavity for the new 1500mah main battery, as well as creating a slot for a secondary battery to power the GPS system. Above this slot, there is an area to mount a protoboard, which will be for the GPS system and for a small screen to mount and read out apogee, post flight. The threaded rod holes were adjusted once again to 3.5mm to reduce lateral play inside the airframe, which worked perfectly.



Figure 17 - V3 sled design

Version 3 addressed all major concerns, finalising this prototyping phase. The next bay will be the final design and will be almost identical to V3, but with cable management holes, external anchor points for battery securing and will be printed from PETG, for its thermal stability.

5.7 - Decoration

The rocket's colour scheme will be a blue body, with a red nose cone and fins. The team name will be written on the side in red along with the team logo. This colour scheme is due to 2 major factors:

- The bright colour will aid with spotting the rocket for recovery purposes.
- The composition of the two contrasting colours makes it visually appealing.

The design will be added using spray paint as well as stencil to add the lettering.

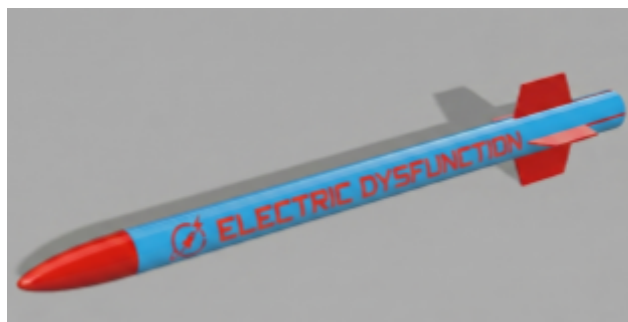


Figure 18 - rocket with decoration

6 - Structural Testing

6.1 - Nose cone drop testing

During the launch, the Rocket is simulated to have an impact speed $\approx 8\text{m/s}$. To assure the nose cone will be strong enough to sustain this impact, it was decided to test the nose cone at a higher speed $\approx 14\text{m/s}$. This was achieved by dropping from an approximate height of 10m. Using:

$$V^2 = U^2 + 2aS$$

Subbing in the height (10m), acceleration due to gravity (9.81 m/s^2) and initial speed (0m/s). The impact speed is shown to be 14-15m/s. The nose cone was to be dropped onto grass to simulate the conditions it will occur under the real launch, however the first attempt landed onto a concrete panel causing a structural failure (**see Figure 19**). After repairing, the nose cone was dropped onto grass with no damage. With this test it is shown that the nose cone will be able to sustain the impact speed without any damage.



Figure 19: Failed nose cone drop test result

6.2 - Compression tests

During the launch, the rocket will undergo a compressive stress due to the acceleration caused by the motor. To ensure the tubing chosen is strong enough to withstand the stress, compressive tests will be performed in the structures lab.

Tubing will be loaded with weights to create a stress which exceeds the maximum compressive stress that the tubing will be submitted to. This will ensure that the tubing won't fail during the flight. To load the tubing, an option is to tie rope to a strong bit of metal which will be placed on the top end of the tube. The rope will be inserted through the centre of the piping. The tube itself would be suspended vertically. Weights will then be attached to the bottom of the rope via a hook until the maximum expected compression force of around 100N is exceeded.

6.3 - Sag test

According to ARCS-CR-2-1-3, the rocket may not sag more than 1% of its total length when suspended from each end. With a length of 88cm, this gives a tolerance of 8.8mm. This will be tested once the lengths of body tubing have been cut - if the current design does not fulfil this requirement, the length of the inner coupler will be increased to provide more resistance to sagging.

6.4 - Fin test

According to ARCS-CR 2-1-6, the fins must be able to sustain a force equal to :

$$F = 2g_{max}M_{fins}$$

$$\text{Where: } g_{max} = \frac{a_{max}}{9.81}$$

Subbing in a_{max} ($107m/s^2$) and M_{fins} (0.043Kg), the required force is 0.94N which is equivalent to a 95.6g mass. To assure the fins are of suitable strength the fins should be tested with a 100g mass. Combining this with the requirement from ARCS-CR-2-1, a mass of 921g was applied equally across the fins to test if they could hold the weight of the rocket without 10 degrees of sag.

From the test shown in **Figure 20**, it was determined that there was negligible deflection of the fins, successfully demonstrating that the rocket fulfils ARCS-CR-2-1.



Figure 20: fin sag testing

As well as the in weight test, an impact test was carried out to determine whether the fins would survive collision with the floor upon descent. Using the testing rig shown in **Figure 21** and dropping it from a height of 10m (simulating an impact speed of 14m/s), it was found that the fin assembly is extremely likely to survive ground impact due to the fins not sustaining any damage from the test.



Figure 21: fin impact testing

7 - Functional testing

7.1 - Parachute, shock cord and eye bolts

The initial shock cord design comprises a 50/50 split between the HMPE and rubber shock cord (see component links). Using a black powder calculation website[3], the shock cord and recovery system would need to withstand 591N. To test this, the



Figure 22 - initial shock cord ratios

shock cord and eye bolts will be put under a force equal to this to see if it can withstand the separation forces. From this, the ratios between HMPE and rubber will be altered to minimise the force applied to the bulkheads whilst also avoiding the shock cord snapping.

For the parachute, a drop test with a simulated weight of the rocket will be carried out from a height of 6 floors (20m). To measure the efficacy of the test, a slow motion camera will be used at the bottom to measure the rate of descent against a known distance - this can then be used to test whether the parachute will allow the rocket to descend at a suitable speed.

7.2 - Separation

Once the inner coupler has been attached to the tubing, separation tests can occur. The fit between the upper and lower sections will be assessed to make sure it is correct. A too tight fit can be adjusted via sanding and a too loose fit can be fixed by adding tape to increase the friction between the coupler and tubing. The rocket should be able to separate via a small pull but not due to gravity alone. The couplers themselves must comply with the statement of requirement ARCS-CR-2-1-7: "All tube couplers must be a minimum of 2 diameters long, with each side of the connector extending a minimum of 1 diameter into its receiving body tube."

The couplers should be tested together by connecting the rocket by the couplers, then adding increasing weights to one side and recording any sag, rotation or movement. Additionally, dropping small weights a small distance, this is to test how the couplers handle sudden loads. Then this process will be repeated for the other side. This will prove that the coupler can withstand the force when being launched to prevent separation.

7.3 - Avionics Testing

Test 1: Smokestopper Test

- Objective: Safely supply battery power without risking catastrophic short circuit or LiPo thermal runaway.
- Method: Battery is connected to the PCB in series via a smokestopper.
- Result: Passed. The smokestopper did not trip, indicating no high-current shorts were present on the board.

Test 2:



Figure 23 - Smoke stopper test

Voltage

Regulation Verification

- Objective: Ensure 8.4V (full battery) is supplied to the PCB and safely stepped down to 3.3V for the logic components and committee payload.
- Method: With the battery connected, a multimeter is used to measure the voltages of the battery connector, buck converter, and committee payload connector.

- Result: Passed. The system took the nominal 8.4V battery input and stepped it down to a stable 3.3V.

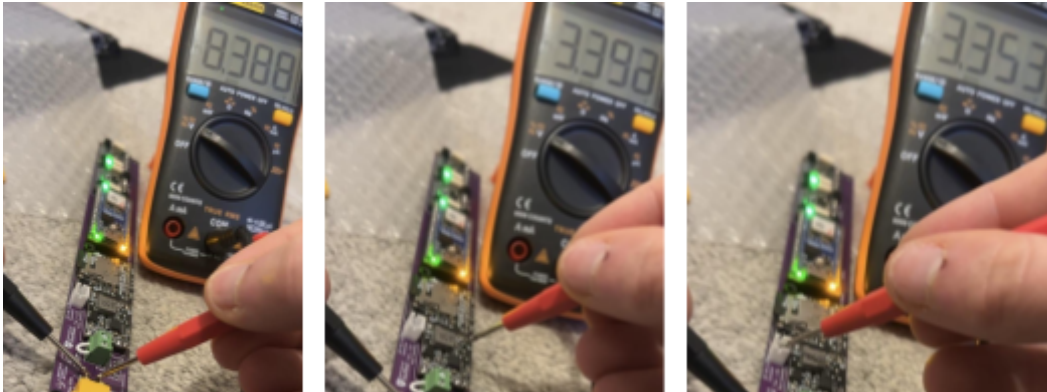


Figure 24 - test 2

Test 3: Functional Verification

- Objective: Verify the ESP32 and all I2C/SPI sensors survived the soldering process and operate correctly.
- Method: The board is powered on and connected to the serial monitor to observe live telemetry.
- Result. Passed. All sensors initialised and logged data to serial monitor at high rates.



Figure 25 - test 3

7.3.1 Tests to be Performed:

Test 4: Vibration and Shock Tolerance

- Objective: Verify that the high-frequency vibrations of the motor burn and the physical shock of parachute deployment will not cause brownouts, loose wires, or the MicroSD card to unseat.
- Method: The avionics sled is fully assembled with the battery secured, powered on and logging. The assembly is then strapped firmly to an orbital sander for 10 seconds, or vigorously shaken by hand, followed by a one meter drop onto a padded surface to simulate deployment shock.
- Expected Result: Passed if the board maintains continuous power, the ESP32 does not reset, and the SD card log shows a continuous stream of data.

Test 5: Simulated Flight Profile (Vacuum Test)

- Objective: Verify that the barometer correctly reads pressure changes, translates them to altitude, and accurately detects the apogee.
- Method: The powered avionics bay is placed inside a vacuum chamber and a partial vacuum is pulled to simulate an ascent to target altitude, and then slowly released to simulate descent.
- Expected Result: Passed if microcontroller logs a smooth, expected altitude curve and correctly flags the apogee.

Test 6: Thermal and Duration Endurance

- Objective: Ensure the system can run for at least an hour when inside an enclosed, unventilated tube.
- Method: The avionics are sealed inside the PETG sled and the outer airframe tube. The system is powered on with the 1500mah main battery and left to log data continuously for a minimum of 1 hour, with 1.5A being drawn from the JST-PH connector.
- Expected Result: Passed if the system logs continuously for 1 hour, the battery drains at an expected rate, and inspection shows no thermal deformation of the PETG or overheating of the buck converter.

8 - Remaining project risks

8.1- parachute not deploying

A potential cause for concern would be the parachute not deploying. This would lead to a catastrophic error if this were to happen as the rocket would fall and pick up far too much speed if it weren't being slowed down by the parachute. This would lead to an impact velocity much higher than that predicted and likely result in the rocket failing and breaking completely.

This can be mitigated by researching methods of packing the parachute when it comes time to assemble it and ensure its been packed well and it will definitely open.

Another concern is the rocket failing to separate, leading to failed parachute deployment. To battle this, extensive fit testing of the couplers will be undergone to assure the correct fit between the upper and lower section of the rocket. A tight fit could prevent the sections from coming apart. Whilst a loose fit could lead to early separation which would highly increase the chance of undergoing damage to the rocket.

8.2 - Eye bolts

Another potential risk would be the force exerted onto the eye bolts being higher than currently expected. For these calculations, the black powder calculator was used to calculate the force exerted on the bulkhead [3].

Should it exceed this, it would lead to the eye bolt getting ripped out of the bulkhead, meaning the parachute wouldn't function and the same thing as the parachute not opening would happen in terms of the high impact speed.

8.3 - Recovery

High apogee and variable weather conditions could result in the rocket becoming invisible to the naked eye. To combat this, a high visibility parachute is used, and an ExpressLRS receiver and u-blox M10 GPS module will communicate with each other, and send GPS coordinates to a Radiomaster Pocket ground station.

8.4 - Total Power Loss

To mitigate the risk of power loss during high vibration events like motor burn and ejection, several hardware and software redundancies have been implemented. A main battery disconnection would power down the microcontroller, risking critical data loss that violates competition guidelines. To prevent this, the main battery is securely mounted to the avionics sled using XT30 connectors, which are accepted in the rocketry community. Additionally, the GPS module operates on an independent backup battery to guarantee location tracking even if the primary system fails. On the software side, flight data is flushed to the microSD card every second to minimize data loss during a power interruption. Through testing it has been verified that if power is temporarily lost and restored, the microcontroller automatically reboots, reinitializes the sensors, and resumes logging in a new flight data file.

9 - Next steps

9.1 - Lessons learned from completed testing

The testing that has been completed so far comprises the nose cone drop test and the fin impact test.

From the nose cone drop test it was concluded that the nose cone did not have to be completely solid to have a suitable structural strength - this led to the current design being modelled as a 8mm thick hollow shell rather than the previous infilled design. Furthermore, as the destruction of the nose cone when dropped on concrete was along the shoulder connection to the head rather than either part by itself, the additional support provided by being glued to the front body tube should provide additional strength and reduce chances of component failure.

From the fin impact test, it was concluded that the 3mm thick plywood fins would be suitable to survive impact at 9.43m/s when glued into place in the rear assembly and attached to the rear body tube. This means that the fin design does not have to be changed drastically for the final design, allowing the team to focus on other areas of the rocket.

9.2 - Future Testing plans

To summarise, the structures testing still to be completed is:

- Body tube compression testing
- Nose cone V2 drop test
- Parachute drop test
- Shock cord force test
- Coupler testing

Body tube compression testing will be done in the structures lab and the tube will rather be loaded with weights to create a stress which exceeds the maximum compressive stress that the tubing will be subjected to (as described in section 6.2) or if available, a section of the tubing will be tested in the UTM's.

V2 of the nose cone will be tested in the same manner as V1 (see section 6.1).

For the parachute, a drop test with a simulated weight of the rocket will be carried out from a height of 6 floors (20m). To measure the efficacy of the test, a slow motion camera will be used at the bottom to measure the rate of descent against a known distance - this can then be used to test whether the parachute will allow the rocket to descend at a suitable speed.

For the shock cord testing, a mass providing a tensile force equivalent to/ greater than that maximum force experienced will be hung from the shock cord to ensure that it does not snap during separation. Another potential option is using the tensile testing machines in the structures lab if available.

To test the coupler a fit test will be carried out during assembly in which it will be examined whether the coupler very slowly slides out when the rocket is inverted therefore indicating a suitable fit.

9.3 - Future Manufacturing

Within the next few weeks, multiple stages of final manufacturing will take place. This includes cutting the body tubing to length and cutting the fin slots using the jig, as well as constructing and inserting the rear assembly that will contain the motor sleeve and the engine mounting brackets. Furthermore, holes will be drilled into the side of the tubing for the launch rails and avionics. For the nose cone, the new design will be 3D printed in the IForge so it can be tested and (if successful) glued into place. Once coupler and sag testing has been completed, the inner coupler will then be glued into position as well as the forward bulkhead and recovery system. With the forward bulkhead then in place, the avionics bay can be fitted into the forward body tube and screwed into place.

On the Avionics side, future manufacturing includes finalizing the physical assembly and integration of the hardware. This begins with completing the 3D-printed avionics bay, which is designed to securely house the main PCB, batteries, and peripheral components. Additionally, the GPS module and apogee display screen will be assembled and mounted to a dedicated perfboard. This perf board will be screwed into the space allocated for it in the avionics bay, above the secondary battery (a 4.35V HiV 1S LiPo). This independent power source ensures the GPS system can continue transmitting coordinates to the ground station even if the main battery disconnects. Once this and some other small tasks such as soldering connectors, mounting and stripping cables, the avionics team will be fully focused on the continued testing of the payload.

10 - References

[1]: Charlesworth, P. (n.d.). *Model Rocket construction*. [online] Model Rocket Construction.doc. Available at:

<https://ukra.org.uk/wp-content/uploads/2024/07/modelrocketconstruction.pdf>.

[2]: Apogee Components, Inc (2020). *Aerotech 29mm Single Use Motor - G78G-10*. [online] Apogeerockets.com. Available at:

https://www.apogeerockets.com/index.php?main_page=product_info&products_id=393 [Accessed 4 Apr. 2026].

[3]: Cook, J. (n.d.). *FFFFg Black Powder Calculator* | *InsaneRocketry*. [online] www.insanerocketry.com. Available at:

<https://www.insanerocketry.com/blackpowder.html> [Accessed Apr. 8AD].